Kawish Hassan

AI/ML Engineer | Agentic AI & Applied Machine Learning
Islamabad, Pakistan | kawish918@gmail.com | +92 331 3724422
linkedin.com/in/kawishhassan | github.com/kawish918 | Portfolio

Summary

Computer Science graduate (Air University, 2026) specializing in applied machine learning, agentic AI, and network security. Build end-to-end AI systems, from agentic SOC systems that reason over live security alerts to multi-agent systems that analyze complex codebases. Experienced in designing system architecture across data pipelines, ML models, APIs, and deployed services. Hands-on experience building production-oriented ML pipelines using autoencoders, LSTMs, and GNNs, and developing agentic systems with LLMs and orchestration frameworks.

Skills

Languages: Python, C++, SQL
ML/AI & Agents: PyTorch, Computer Vision (CNNs, ResNet), Graph Neural Networks, LLMs, Multi-Agent Systems (LangGraph, LangChain, ReACT), Hugging Face, Scikit-learn, Local LLM Deployment (Ollama), Vector Databases (Pinecone, ChromaDB)
Tools & Databases: Docker, Linux, Git, CUDA, PostgreSQL, Firebase, Supabase
Development: FastAPI, REST APIs, React, PySide6, System Architecture

Experience

Federal Board of Revenue (FBR) Jun 2025 – Aug 2025
AI Intern *Multan, Pakistan*

- Designed and built a Tax Analysis System that automatically generates complex data queries to surface concealed revenue and tax loopholes from large government datasets.
- Streamlined the manual query-writing process for auditors, accelerating data-driven, evidence-backed tax audits.

Projects

SentiNet – Adaptive AI-Based Network Intrusion Detection & Prevention System Final Year Project

- Architected a full-stack IDS/IPS platform: a FastAPI central server, a React admin dashboard, and a Windows endpoint agent for real-time network monitoring and automated threat response.
- Built an adaptive multi-model detection engine – Autoencoder, LSTM, and Graph Neural Network – to flag anomalous network traffic and counter concept drift as attack patterns evolve.
- Developed a ReACT-based SOC agent on Llama-3.1-8B-Instruct that reasons over live alerts, pulls supporting context, and recommends or executes response actions autonomously.
- *Stack: Python, PyTorch, FastAPI, React, PostgreSQL*

Person Re-Identification System Personal Project

- Built a person re-identification pipeline in PyTorch with a ResNet-18 backbone for cross-camera identity matching.

E-GAT – Graph Neural Network for Network Intrusion Detection Research Project

- Built a graph neural network-based IDS that models network traffic as a graph, folding flow-level features directly into attention and classification instead of scoring each flow in isolation.
- Reached 98.63% accuracy with zero failed attack classes across 15 attack types on a highly imbalanced 3.1M-flow dataset – attention alone lifted accuracy 49% over a graph-convolution baseline.

MediGuard – Agentic Clinical Observability System Personal Project

- Built a multi-agent medical command center on MedGemma, MedASR, and MedSigLIP that autonomously drafts, verifies, and safeguards clinical documentation.

codeGRAPH – Self-Hosted Multi-Agent Codebase Analyzer Personal Project

- Built a self-hosted, privacy-first codebase visualizer: a LangGraph-orchestrated agent swarm parses every file in a repository in parallel and builds an interactive dependency graph, answering architecture questions through fully local inference via Ollama.

Education

Air University 2022 – 2026
Bachelor of Science in Computer Science

Certifications

Machine Learning Specialization – DeepLearning.AI
AI for Cybersecurity Specialization – Johns Hopkins University
Advanced Learning Algorithms – DeepLearning.AI
Building and Training Neural Networks with PyTorch – Packt
Python for Data Science, AI & Development – IBM